

LLaMA, PaLM

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Introduction

- **LLaMA (Large Language Model Meta AI)** is a family of **Large Language Models (LLMs)** developed by **Meta AI** (formerly Facebook AI Research).
- It is based on the **Transformer architecture** and is designed to perform a wide variety of Natural Language Processing (NLP) tasks such as text generation, question answering, summarization, translation, code generation, and conversational AI.
- The first LLaMA model was introduced in **February 2023** to provide researchers with efficient, high-performing foundation models trained primarily on publicly available datasets.
- Later versions expanded commercial availability, multilingual capabilities, larger context windows, and multimodal features.

- **LLaMA = Large Language Model Meta AI**
- **Large Language Model** – A deep learning model trained on massive amounts of text.
- **Meta AI** – The Artificial Intelligence research division of Meta.

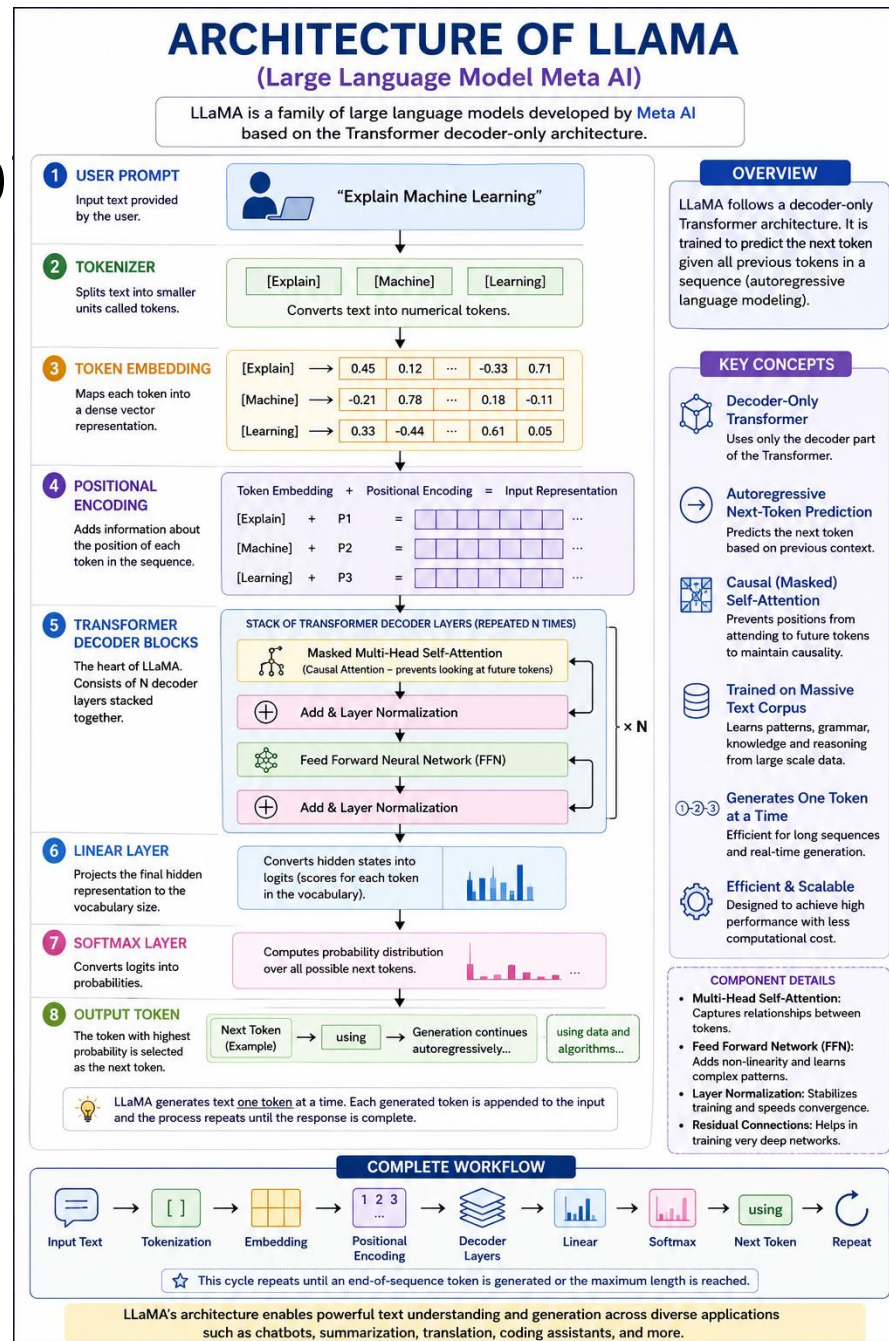
Why was LLaMA Developed?

- Meta developed LLaMA to:
- Democratize AI research through openly available model weights (subject to licensing).
- Build efficient models that require fewer computational resources than many earlier LLMs.
- Enable researchers and developers to fine-tune models for domain-specific applications.
- Accelerate innovation in open AI ecosystems.

Evolution of LLaMA

Version	Release Year	Main Features
LLaMA 1	2023	Research-oriented, efficient Transformer models (7B–65B)
LLaMA 2	2023	Commercial license, improved performance and safety
LLaMA 3	2024	Better reasoning, coding, multilingual support
LLaMA 3.1	2024	405B model, 128K context window, stronger tool use
LLaMA 4	2025	Native multimodal models (text + images), very large context windows, improved efficiency

Architecture of



How LLaMA Works

Step 1: Input

User enters:
"Artificial Intelligence is"

Step 2: Tokenization

Sentence becomes tokens:
Artificial Intelligence is
↓
[1456] [2879] [412]

Step 3: Embedding

Each token is converted into a dense numerical vector.
Example:
Artificial ↓ [0.41 0.12 0.78 0.34 ...] Intelligence ↓
[0.91 0.56 0.20 0.63 ...]

Step 4: Positional Encoding

Position information is added.
Artificial (Position 1) Intelligence (Position 2) is (Position 3)

Step 5: Transformer Decoder

Each decoder layer performs:
Self Attention
Feed Forward Network
Residual Connections
Layer Normalization

Step 6: Prediction

Model predicts:
Artificial Intelligence is ↓ transforming
Then:
Artificial Intelligence is transforming ↓ the
This continues until the response is complete.

7. Training Process

Phase 1: Pre-training

LLaMA is trained on trillions of text tokens using **self-supervised learning**, where the model learns to predict the next token from previous tokens.

Example:

Input:

Machine Learning is a branch of _____

Target:

Artificial Intelligence

Phase 2: Fine-tuning

The model is adapted for specific tasks such as:

Chatbots

Coding assistants

Medical assistants

Financial assistants

Educational tutors

Key Features

- Open-weight model family
- Transformer decoder architecture
- Strong multilingual support
- Long-context processing
- Code generation
- Mathematical reasoning
- Tool calling and agent support
- Fine-tuning capability
- Efficient deployment across different hardware

Applications

Education

- Intelligent tutoring
- Question answering
- Automatic note generation

Healthcare

- Medical report summarization
- Clinical decision support
- Medical chatbots

Software Development

- Code generation
- Code explanation
- Bug fixing

Business

- Customer support
- Email drafting
- Report generation

Research

- Literature summarization
- Research assistance
- Information retrieval

Finance

- Financial document analysis
- Investment report generation

Advantages

- Open-weight ecosystem
- High-quality text generation
- Strong coding performance
- Flexible fine-tuning
- Local deployment options for smaller models
- Large developer community
- Cost-effective compared with many proprietary models

Limitations

- Can generate incorrect or fabricated information (hallucinations)
- Requires significant computing resources for large models
- Performance depends on prompt quality
- May reflect biases present in training data
- Needs safety guardrails for production systems

PaLM (Pathways Language Model)

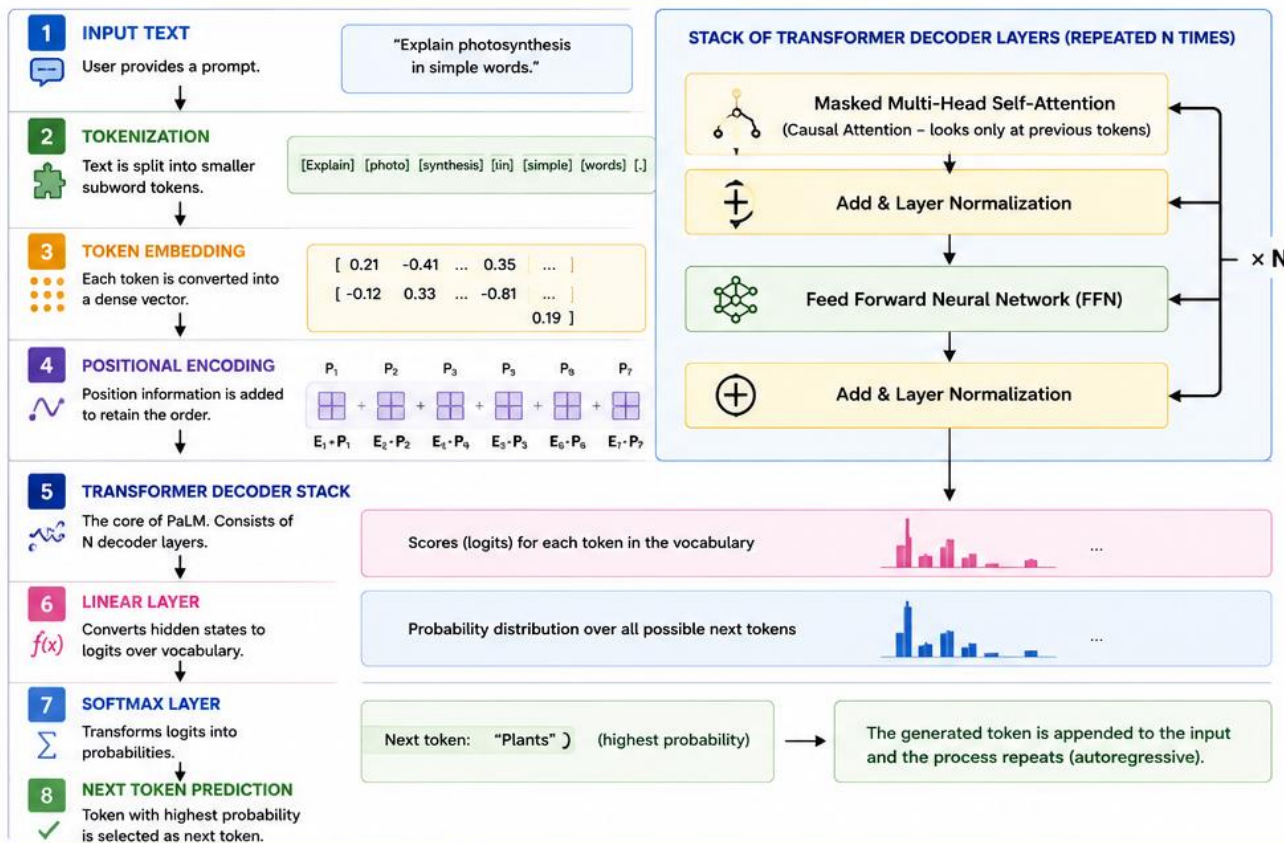
- **PaLM (Pathways Language Model)** is a family of **Large Language Models (LLMs)** developed by **Google AI**.
- It is based on the **Transformer architecture** and is designed to understand, generate, summarize, translate, and reason over natural language.
- PaLM was introduced in **2022** as one of the largest and most capable language models of its time.
- Unlike traditional AI systems trained for a single task, PaLM is a **foundation model** trained on massive amounts of text and code, enabling it to perform a wide variety of tasks with minimal additional training.



PALM ARCHITECTURE

(Pathways Language Model)

PaLM is a large language model developed by Google AI based on the Transformer decoder architecture and the Pathways system for efficiency and scalability.



What is PaLM?

PaLM (Pathways Language Model) is a family of large language models trained by Google AI. It uses the Pathways system which enables training of very large models efficiently across many tasks and modalities.

HOW PaLM WORKS (STEP-BY-STEP)

- 1 The user enters a prompt.
- 2 The text is tokenized into smaller pieces.
- 3 Tokens are converted into dense vectors (embeddings).
- 4 Positional encodings are added to represent the order.
- 5 The sequence passes through N decoder layers. Each layer applies masked self-attention, adds & normalizes, then applies FFN, and adds & normalizes again.
- 6 The final hidden representation is projected to vocabulary size using a linear layer.
- 7 Softmax converts scores into probabilities.
- 8 The token with the highest probability is chosen as the next token and the process continues.

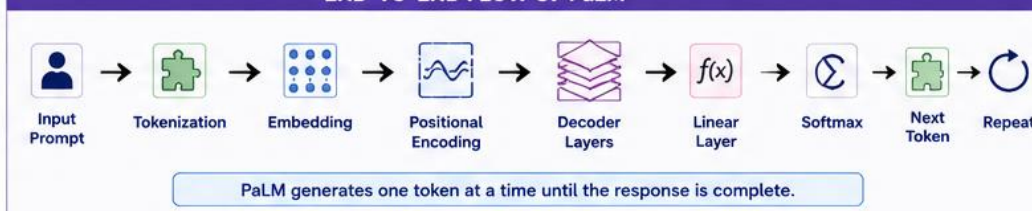
KEY ARCHITECTURAL HIGHLIGHTS

- Decoder-Only Transformer**
PaLM uses a decoder-only architecture similar to GPT models.
- Causal (Masked) Self-Attention**
Each position attends only to previous positions to preserve autoregressive property.
- Pathways System**
Enables training a single model on diverse tasks and datasets efficiently at very large scale.
- Scalability**
Designed to scale to hundreds of billions of parameters (540B in PaLM).
- Multilingual & Multi-task**
Trained on multilingual data and capable of solving multiple tasks with few-shot/zero-shot learning.

MODEL SPECIFICATIONS (PaLM)

Parameters	540 Billion
Layers	118 Decoder Layers
Hidden Size	18,432
Attention Heads	144
Vocabulary Size	~250,000
Training Data	Multilingual Text, Code, Web, Books,

END-TO-END FLOW OF PaLM



EXAMPLE

Prompt: What is the capital of France?

PaLM Output: The capital of France is Paris.

(Generated token by token)



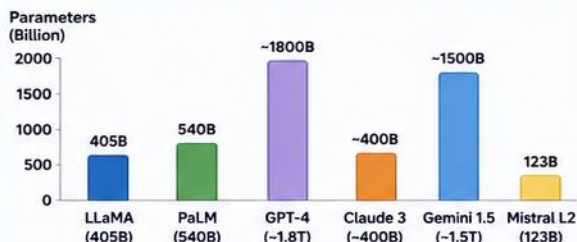
COMPARISON OF MAJOR LLMs

LLaMA vs PaLM vs Other Leading LLMs

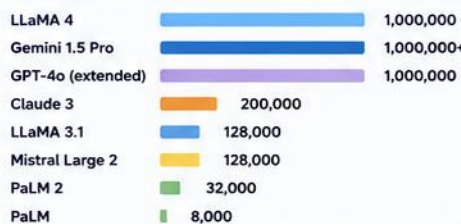


FEATURE	 LLaMA (Meta AI)	 PaLM (Google AI)	 GPT-4 / 4o (OpenAI)	 Claude 3 (Anthropic)	 Gemini 1.5 (Google AI)	 Mistral Large 2 (Mistral AI)
 Developer	Meta AI	Google AI	OpenAI	Anthropic	Google AI	Mistral AI
 First Released	2023 (LLaMA 1)	2022	2023 (GPT-4) 2024 (GPT-4o)	2024 (Claude 3)	2024	2024
 Model Family	LLaMA 1, 2, 3, 3.1, 4 (latest)	PaLM, PaLM 2	GPT-4, GPT-4o	Claude 3 (Haiku, Sonnet, Opus)	Gemini 1.0, 1.5	Mistral Large, Mixtral, etc.
 Architecture	Transformer (Decoder-only)	Transformer (Decoder-only + Pathways)	Transformer (Decoder-only)	Transformer (Decoder-only)	Mixture-of-Experts + Transformer	Mixture-of-Experts + Transformer
 Parameters (Largest Model)	Up to 405B	540B	Up to ~1.8T (GPT-4) (Exact not public)	Up to ~400B (Claude 3 Opus)	Up to ~1.5T (Gemini 1.5 Pro)	Up to 123B (Mistral Large 2)
 Training Data (Approx.)	15+ Trillion tokens	780+ Billion tokens	13+ Trillion tokens	Varies (large proprietary dataset)	Trillions of tokens (Multimodal)	Trillions of tokens
 Context Window (Largest)	128K (LLaMA 3.1) Up to 1M (LLaMA 4)	8K (PaLM) 32K (PaLM 2)	128K (GPT-4o) Up to 1M (extended)	200K (Claude 3)	1M+ (Gemini 1.5 Pro)	128K (Mistral Large 2)
 Open Weights	✓ Yes (Research license)	✗ No	✗ No	✗ No	✗ No	✓ Yes
 Multimodal Capabilities	Yes (LLaMA 4: Text, Image, Video)	Limited (mainly text) (vision via PaLM-E)	Yes (GPT-4o: Text, Image, Audio, Video)	Yes (Text, Image)	Yes (Text, Image, Audio, Video)	Yes (Text, Image)
 Strengths	Open, flexible, good reasoning, long context, cost-effective	Strong reasoning, multilingual, math & code	Very strong general ability, tool use, big ecosystem	Safe, reliable, long context, ethical focus	Multimodal power, very long context, integration with Google	Efficient, high performance, open models
 Best For	Research, customization, local deployment, enterprise use	Enterprise, research, multilingual tasks, reasoning	General purpose, chatbots, coding, tool use, automation	Enterprise, documents, analysis, responsible AI applications	Large-context apps, multimodal AI, Google ecosystem	Edge deployment, enterprises, cost sensitive use
 Availability	Download from Meta (Hugging Face)	Via Google Cloud API	Via OpenAI API (ChatGPT)	Via Anthropic API (Claude.ai)	Via Google AI Studio / Vertex AI	Download from Mistral AI / Hugging Face

PARAMETER SCALE COMPARISON



CONTEXT WINDOW COMPARISON



OPEN WEIGHTS?

LLaMA	✓ Yes
PaLM	✗ No
GPT-4 / 4o	✗ No
Claude 3	✗ No
Gemini 1.5	✗ No
Mistral Large 2	✓ Yes

KEY TAKEAWAYS

-  LLaMA and Mistral offer open weights for flexibility and local use.
-  GPT-4o and Gemini 1.5 lead in multimodality and ecosystem integration.
-  Claude 3 focuses on safety, reliability, and long context handling.
-  PaLM was a pioneer with strong reasoning and multilingual capabilities.
-  Choose the model based on your use case, budget, privacy and deployment needs.